

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

23rd May 2014

Subject CT2 – Finance and Financial Reporting

Time allowed: Three Hours (10.30 – 13.30 Hrs.)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** Which of the following financial ratios will be affected by an error in recording the value of stock in the financial statement?
- i)** Inventory turnover ratio
 - ii)** Current ratio
 - iii)** Earnings per share
 - iv)** Interest cover ratio
- A)** Option i) only
B) Option i) and ii)
C) Option i), ii) and iii)
D) All the above [2]
- Q. 2)** Which of the following will not be true if the company increases its gearing whilst keeping the total capital employed in the business constant and the company does not pay out dividends (assume other things remain constant)?
- i)** The tax liability of the company will reduce
 - ii)** Equity shareholders will expect company to earn higher returns on capital employed
 - iii)** Asset utilisation ratio will remain unchanged as the amount of equity capital will reduce by an amount equal to the amount of debt raised (assuming that equity shares are bought back at par and debt is also issued at par)
- A)** Option i) and ii)
B) Option ii)
C) Option ii) and iii)
D) Option iii) [2]
- Q. 3)** Which of the following is a tax deductible expense from business income?
- i)** Cost of machinery purchased during the year
 - ii)** Interest paid on loan taken to acquire machinery
 - iii)** Training expenses incurred to train employees on how to use the new machine
- A)** Option i) and ii)
B) Option ii) only
C) All the above
D) Option ii) and iii) [2]

Q. 4) Which of the following is not suitable to hedge the risk of rise in dollar rates for an importer with varying amounts of dollar outflows each month?

- i)** Currency swap
- ii)** Selling forward contracts on dollar maturing in various months for expected dollar outflows
- iii)** Buying futures on dollar maturing in various months for expected dollar outflows

- A)** All the above
- B)** Option i) and ii)
- C)** Option ii) and iii)
- D)** Option i) and iii)

[2]

Q. 5) Which of the following statements are False:-

- i)** Junk bond is a bond which meets the usual requirements of income cover but does not meet the requirement of capital cover for institutional investors.
- ii)** Eurobond is an international bond issued which can only be issued by a government. The bonds are usually traded in the traditional bond markets.
- iii)** Floating-rate note is a type of Eurobond with a variable rate of interest.
- iv)** Bulldog is a sterling denominated foreign bond issued by an overseas borrower in the traditional UK bond market.

- A)** Option iii) only
- B)** Option i) and ii)
- C)** Option i), ii) and iv)
- D)** Option iii) and iv)

[2]

Q. 6) Which of the following statements are False:-

- i)** Clean price is the price of a bond with allowance for accrued interest.
- ii)** Exercise price is the price at which an underlying security can be sold to (for a put) or purchased from (for a call) the writer or issuer of an option
- iii)** Bid price is the price at which a market maker offers to sell a security.
- iv)** Callable bond is a bond containing provisions that allow the holder to sell it back at a predetermined price at certain times during its life.

- A)** Option ii) only
- B)** Option iii) and iv)
- C)** Option i), iii) and iv)
- D)** Option i) and iii)

[2]

Q. 7) Which of the following statements correctly defines the two basic issues of finance and investment decisions:-

- i) What real assets should the firm invest in?
- ii) How should the cash for the investment be raised?

- A) Option i) is Investment Decision and Option ii) is Finance Decision
- B) Option i) is Finance Decision and Option ii) is Investment Decision
- C) Both Option i) and ii) are Investment Decisions
- D) Both Option i) and ii) are Finance Decisions

[2]

Q. 8) Which of the following is not a component of Agency Cost:-

- i) Cost incurred as commissions/brokerage paid to Agents
- ii) Cost incurred to monitor the managers
- iii) Cost incurred in seeking to influence the actions of managers.
- iv) Cost incurred because the managers do not act in the owners' best interests

- A) Option i) only
- B) Option iii) and iv)
- C) Option ii), iii) and iv)
- D) Option ii) and iii)

[2]

Q. 9) A unit trust currently comprising 50,000 units has total assets of Rs.10,000,000 valued at mid-market prices. The current cost of buying the assets held by the trust using the price of the underlying assets and allowing for dealing costs (commission and stamp duty) would be Rs.10,500,000. If the assets were sold, they would realize Rs.9,850,000. Assume a 8% bid-offer spread.

What is the effective spread for a person who is buying when the trust is expanding and selling when the trust is contracting?

- A) 8.00%
- B) 12.38%
- C) 13.70%
- D) 9.38%

[2]

- Q. 10)** A unit trust currently comprising 2000 units has total assets of Rs.150,000 valued at mid-market prices. The current cost of buying the assets held by the trust using the price of the underlying assets and allowing for dealing costs (commission and stamp duty) would be Rs.152,000. If the assets were sold, they would realize Rs.145,000. Assuming a 6% bid-offer spread.

What is the effective spread for a person who is buying when trust is contracting and selling when trust is expanding?

- A)** 6.00%
- B)** 7.24%
- C)** 10.33%
- D)** 1.46%

[2]

- Q. 11)** i) Define the following ratios and state its use:

a) Current ratio (2)

b) Debtors turnover period (2)

- ii) Xylo Ltd deals in goods with a high inventory turnover period. Its finance manager wishes to ascertain whether the company would be able to meet its short term liabilities. Explain which ratio should he track on a regular basis and state the equation for the ratio suggested. (1)

[5]

- Q. 12)** i) List the major type of business entities and differentiate their characteristics with relation to ownership, liability, and legal status. (4)

- ii) An individual is contemplating opening a business trading in certain goods. He has short listed two options, either to start as a Sole Trader or float a limited company with the help of his friends. Only from the point of view of minimizing the tax amount, suggest, with appropriate reasons, a form of business entity he should opt for. (2)

- iii) Assuming that the individual chooses to start with a limited company in a country with corporation tax rate of 50% and a dividend distribution tax rate of 20%. It has earned profit before tax of Rs.100000 in the first year. The company expects to earn 15% per annum before tax on retained capital. The individual has alternative option of investment where he/she can earn 7.25% per annum tax-free. What should be the Dividend policy of the company?

The individual has no surplus funds to invest however he/she has other source of income to sustain his/her daily expenses.

- A)** Distribute Rs.100000 as dividend
- B)** Do not distribute any dividend
- C)** Distribute Rs. 50000 as dividend
- D)** Distribute Rs. 30000 as dividend

(2)

[8]

- Q. 13)** The following balances have been extracted from the books of JK Ltd. a manufacturing entity, for the financial year ended on 31st March 2014.

Balances as on 31st March 2014

(Amt in INR Million)

	Dr.	Cr.
Equity share capital		1,000
Purchases	5,600	
Sales		10,250
Trade receivables	4,400	
Trade payables		3,200
Sales returns	125	
Purchase returns		50
Administrative expenses	230	
Advertisement and marketing expenses	560	
Octroi charges	5	
Freight inward	10	
Plant and machinery	770	
Land	150	
Factory building	250	
Motor vehicles	60	
Cash in hand and at bank	141	
Bank loan		662
Salaries	365	
Wages	1,700	
Power and fuel	1,060	
Auditor's fees	5	
Insurance	45	
Legal consultancy fees	5	
Interest on loan	66	
Directors' remuneration	180	
Retained earnings		1,536
Opening inventory	970	
Printing and stationery	5	
Insurance claim received		4
Total	16,702	16,702

Additional Information:

- Depreciation to be provided as follows on reducing balance:
Building- 5% p.a.
Plant and machinery- 15% p.a.
Motor vehicles (used by staff & directors) - 20% p.a.
- Goods worth INR 10 million and having sales value of INR 12.50 million were given to M/s. DP & Sons, one of our distributors, as free sample. The accountant wrongly booked it as credit sales.

3. Considering the past experience about bad debts, the company has decided to make a provision for doubtful debts at 5 % of trade receivables.
4. A fire occurred in the godown on 10th March 2014 but was doused before it could cause significant damage. However, it burnt goods worth INR 5 million. The insurance company admitted a claim of only INR 4 million and acceptance of claim intimated on 25th March 2014. The payment was received on 31st March 2014 and was booked under the head "Insurance claim received"
5. Physical counting of stock in godowns and at the manufacturing unit as on 31st March 2014 revealed that the stock was worth Rs. 1315 million (based on our manufacturing cost).

As the company's chief accountant, you have been requested to prepare the following after considering additional information provided:

i) Prepare an income statement for the year ended on 31st March 2014. (12)

ii) Prepare a statement of financial position as on 31st March 2014. (10)

The financial statements should be in a form suitable for publication in so far as it is possible from the information provided. [22]

State assumptions if any

- Q. 14)**
- i) Picadelly Ltd. is a private company in United Kingdom and it wants to have its shares listed on the London Stock Exchange. However it doesn't currently wish to raise additional capital. It has sought your help to identify the easiest way to achieve this. Besides it also wants to understand if the method identified has any pre-requisites in terms of its current shareholding pattern. Draft a reply to the company. (2)
 - ii) Based on your great advice and service offered to Picadelly Ltd., Buckingham Ltd which is currently dabbling with the idea of raising capital through a fresh equity issue, approaches you to understand the benefits of underwriting. Draft a reply to the company. (4)
 - iii) For each of the following clients, please advise the best method(s) for raising equity capital along with proper reasoning and explanation
 - a) A biotechnology research start-up (2)
 - b) An established, well reputed and profitable company (3.5)
 - c) A medium sized company which is unsure of its share value (2)
 - d) Government owned company wherein government is seeking to disinvest (1.5)
- [15]

- Q. 15)** ABC Ltd. is a fully equity capitalized company. It is listed on the Sensex and its price at end of each month is benchmarked below against the Sensex levels for past 1 year. A 9% 10 year Central Government bond (Rs. 100 par value) closed at Rs. 106.71 at day end yesterday.

Calculate the cost of capital of ABC Ltd. as on today.

Months	Sensex returns	Returns on ABC Ltd. Shares
1	0.750%	4.360%
2	0.667%	1.000%
3	1.000%	4.000%
4	1.333%	4.000%
5	1.500%	-2.000%
6	0.583%	5.000%
7	0.833%	1.500%
8	-0.417%	-4.000%
9	0.500%	1.000%
10	1.000%	2.000%
11	1.417%	2.000%
12	0.833%	1.500%

[12]

- Q. 16)** A prominent dealer of Audi Vehicles has brought out a new scheme of Operating Lease for New Vehicles. In this scheme the dealer provides the right to use of the new vehicle for 3 years to the lessee for lease payments paid at the end of each month for the next 3 years. There is no transfer of ownership. At the end of 3 years, the dealer takes back the possession of the vehicle.

XYZ Ltd. is considering opting for the Operating Lease for its CEO. Other options it is considering are to buy the Vehicle outright or to finance it by a loan. It is assumed that it will sell the car at the end of 3 years.

Suggest the best option to minimize the overall cost to the company considering following facts.

Outright Purchase Price is Rs. 30 Lakhs

Lease Payment per month is Rs. 74,000

Loan is available @18%, Down Payment of 10% of price and 3 installments payable at the end of each year.

Estimated sale value of the vehicle at the end of 3 years is Rs. 15 Lakhs

Company can claim 15% depreciation on the car on written down basis.

Assume Tax rate at 35%.

Appropriate interest rate for discounting is 12%

Hint - Consider tax deduction of interest paid, rental paid, capital loss for sale of vehicle.

[18]
